Smart Energy Conservation Game

Project 73 final report. Planning a household's electricity day against a fixed budget, in real time.

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Abstract

We built a browser game in which a player runs one household day against a fixed energy budget. Ten appliances are switched on and off in real time while the screen shows energy, CO_2 and cost. This report covers the game, the data behind it and what that data shows. We collected appliance power figures from 3 published sources (33 rows) and compared them with the game's values: 7 of 10 fall inside the published ranges. No human play data exists yet, so the analysis uses 612 days generated by six scripted household strategies. On a Medium day (10 kWh budget), meeting the budget cut energy by 56% against a full-comfort day and kept 78% of the comfort the household wanted. Moving flexible use out of the evening peak cut cost by a further 32% (95% interval 30 to 33) at identical energy and comfort. All of this rests on simulated players.

1. Problem and objectives

A household meets its electricity use as a monthly bill, long after the choices behind it. Nobody sees what one hour of air conditioning cost while there is still time to act, and few people know which appliances matter. The brief asks for a game with a fixed daily energy budget, appliances to operate, and a visible score, energy consumed, CO_2 equivalent and cost. Our objectives were to meet that brief, to ground the appliance numbers in published data, to find out from the data which habits save the most, and to describe how the game could be used for sustainability teaching.

2. The game

A day runs from 00:00 to 24:00 at ten real seconds per game hour (four minutes; 2× and 4× speeds exist). The player clicks appliances in an animated cutaway house to switch them on or off. The fridge starts on. Live panels show a budget ring, power now, CO_2, cost, comfort and an hourly chart. At 24:00 the player gets a score, a medal, a breakdown, tips built from their own day, and a comparison with their previous run on that level. CO_2 is also shown as kilometres driven, using the EPA figure of about 393 g per mile for a typical passenger car [2].

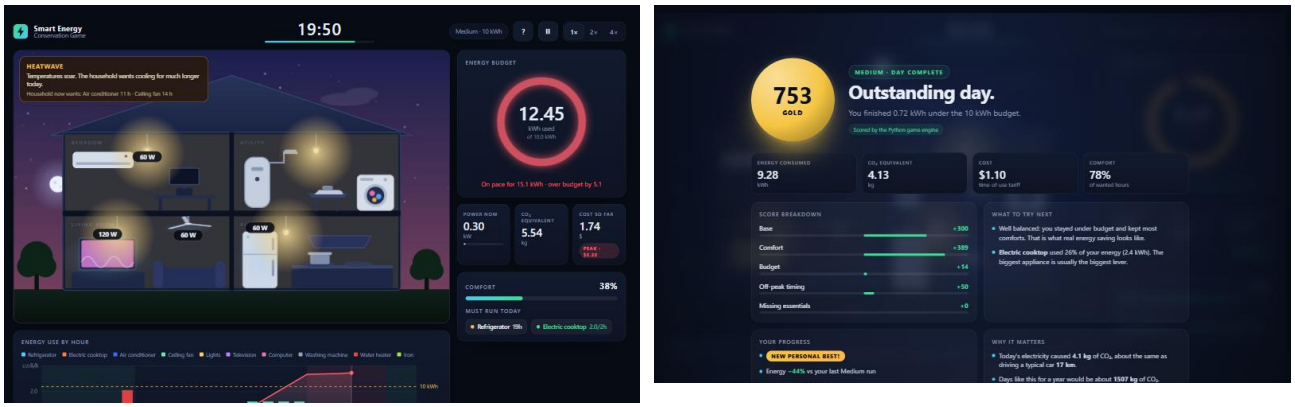


Figure 1. Left: a Medium heatwave day at 19:50, over budget (red ring) with peak pricing active. Right: the results screen for a day that finished under budget.

Levels have budgets of 14 kWh (Easy), 10 kWh (Medium) and 7 kWh (Hard). Five random events change what the household wants that day: a heatwave (air conditioner 11 h, fan 14 h), guests (cooktop 3 h, television 7 h, lights 9 h), a work-from-home day, a laundry day and a cold snap (water heater 3 h). Prices depend on the hour: 0.08 per kWh off-peak (22:00 to 06:00), 0.15 standard, 0.30 at peak (17:00 to 22:00). CO_2 is energy times 0.445 kg/kWh, the 2024 IEA global average [1].

Appliance	Power (W)	Hours wanted	Comfort weight	Essential
Refrigerator	60	24	2.0	yes
Electric cooktop	1200	2	2.5	yes
Air conditioner	1500	8	3.0	
Ceiling fan	60	10	1.0	
Lights	60	6	2.0	
Television	120	4	1.5	
Computer	100	5	1.5	
Washing machine	500	1	1.5	
Water heater	2000	1	2.0	
Iron	1000	1	1.0	

Table 1. The ten game appliances. Running all of them for their wanted hours takes about 21.3 kWh, so every level forces trade-offs.

The score is calculated as follows and never goes below zero.

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score = 300 + 500 × comfort + budget + 100 × (off-peak share − peak share) − 200 × (essentials missed)
budget = 200 × (B − used)/B when under budget, and −600 × (used − B)/B when over, with B the daily budget
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Comfort is the weighted share of wanted hours provided, counted over the non-essential appliances only, so running just the fridge and cooktop earns nothing. The rating (Gold, Silver, Bronze) compares the score with the best score a good plan reaches for that level and event, called par: Gold at 90% of par or more, Silver at 78%, Bronze at 65%.

The front end (HTML, CSS, JavaScript, SVG and canvas) runs the live simulation inside a Streamlit page. Python holds the rules and calculates the official score when the day ends. The browser calculates the same score live; a test compares the two on random days for every level and event.

3. Data: collection and cleaning

3.1 Appliance power figures

Raw table: 33 rows copied from three pages (EnergySage, RunWatts and Enphase), each with its URL, page date, the appliance name as printed, the low and high value and the unit [3][4][5]. We also tried four U.S. Department of Energy pages; all returned errors, so nothing was taken from them. Cleaning removed 0 duplicates and 0 impossible ranges, converted 1 yearly energy figure (a fridge's 400 to 800 kWh a year) to average watts, and set aside 7 rows the game does not use (standby, idle, other air-conditioner types). Each remaining row was mapped to a game appliance and compared with the game's value on the same basis (a fridge as a 24-hour average, everything else as power while running).

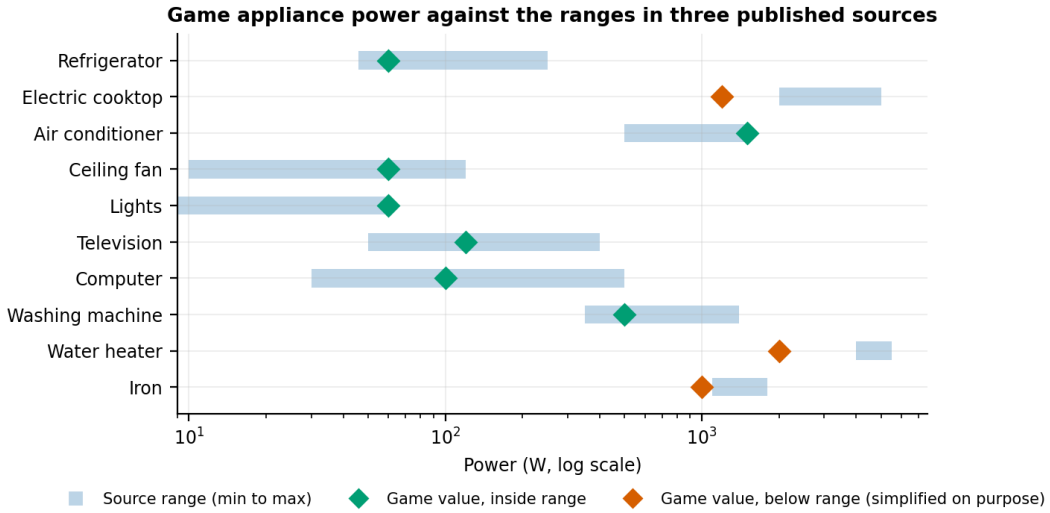


Figure 2. Game values against published ranges. 7 of 10 are inside. The cooktop (one burner against whole-oven figures), water heater (a small unit so one hour is playable) and iron (1,000 W against 1,100 to 1,800 W) are below, and the first two are deliberate simplifications.

The sources disagree with each other in places. For the fridge, the yearly energy figure implies 46 to 91 W on average, while another page gives 150 to 250 W as typical average use. The game's 60 W sits in the first range. We kept the disagreement in the table and did not pick a winner.

3.2 Play logs

Each finished day is one row of 51 columns: player id, level, event, budget, par, score, rating, energy, CO_2, cost, comfort, off-peak and peak share, missing essentials, hours per appliance and energy per hour. The game can export a player's saved runs as an anonymous CSV (random id, no names), and a cleaning script merges any such files with the simulated ones. Its rules drop incomplete rows, duplicate run ids, unknown levels or events, impossible values and rows whose CO_2 or hourly energy does not add up to the total. The merge of 612 rows removed none, because simulated rows are generated clean; the rules are exercised by tests with deliberately broken rows and by a real browser export that passed through them.

No human play data has been collected yet (0 runs). Every result below is from 612 simulated days, and the scripted players are our own invention. They are a controlled way to see how different habits score under the game's rules. They say nothing about how people actually play.

Strategy	What it does	Runs
Full comfort	Every appliance for its wanted hours at natural times; ignores the budget	144
Casual	Natural times, but a random 30% to 110% of the wanted hours	144
Budget cutter	Full comfort, then removes the hours that cost least comfort per kWh until the budget is met	144
Peak-aware	Chooses the cheapest hours each appliance can reasonably run in, then trims to the budget, peak hours first	144
Optimizer	Improves the peak-aware plan by local search (the method that sets par)	18
Essentials only	Fridge and cooktop and nothing else	18

Table 2. The six scripted strategies, run on every level and event (eight seeds each, one for the last two).

4. Par scores

A rating means little unless par is right. We estimate par by searching for the best day on each level and event. Our first search started from an empty day and got stuck on some event days: a simple scripted plan beat it in 115 of 612 runs, by up to 21% (Easy with guests: 642 against 777). We rebuilt the search to start from the scripted plans as well, recomputed all 18 pars, and checked again: no strategy now beats par on any level or event. Par is still an estimate from a random search, accurate to about 2%.

5. Results and inferences

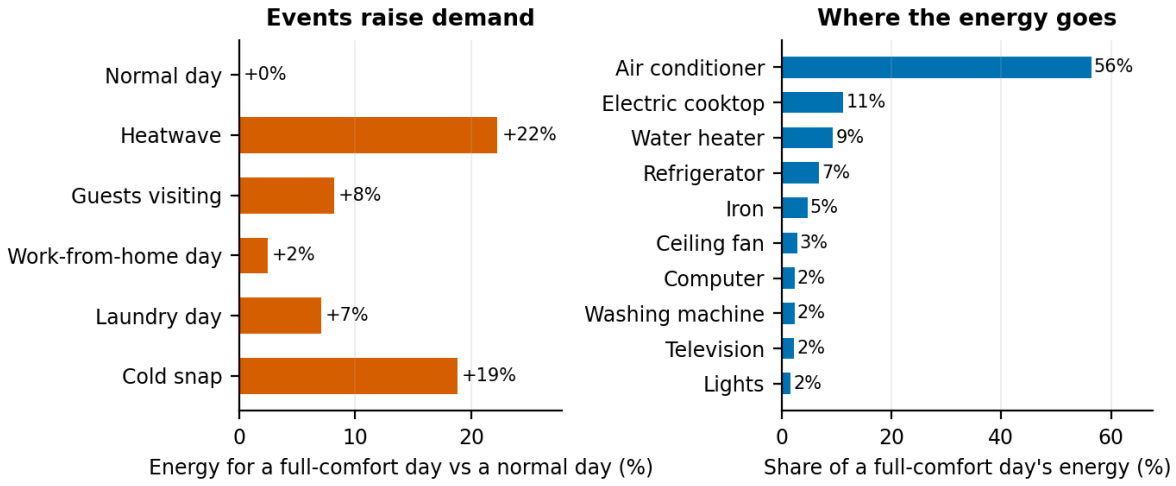


Figure 3. Left: energy needed for a full-comfort day on each event day compared with a normal day. Right: share of a full-comfort day's energy by appliance.

The air conditioner alone takes 56% of a full-comfort day. With the cooktop (11%), water heater (9%) and iron (5%) it makes 82%. Lights, television, computer, fan and washing machine are 11% together. A player who spends effort on the small loads is working on the wrong end. Events move demand by 2% to 22% (heatwave 22%, cold snap 19%), which is why par is set per event.

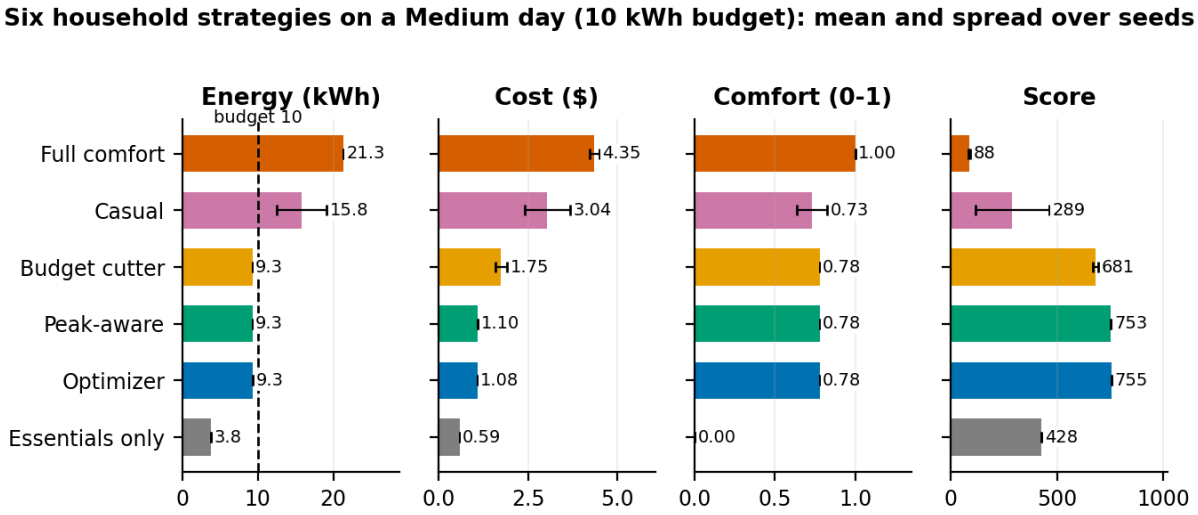


Figure 4. The six strategies on a Medium day without an event. Bars are means over seeds and whiskers show one standard deviation.

Staying within budget costs comfort. A full-comfort day uses 21.3 kWh and is over budget in 100% of runs. The budget cutter and peak-aware strategies use 9.3 kWh, which is 56% less, and keep 78% of the wanted comfort. Casual play, with no plan, is over budget in 97% of runs. The five strategies that provide comfort show a correlation of r = 0.79 between energy and comfort (the runs are not independent, so we report the figure as description only).

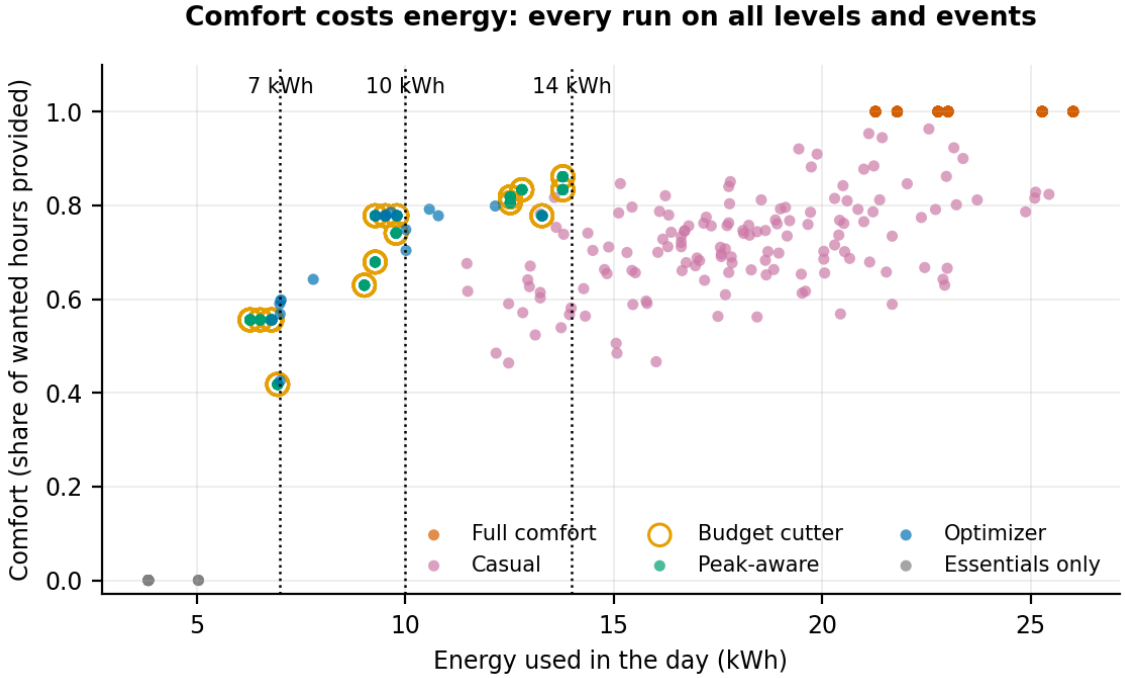


Figure 5. Every simulated run. Peak-aware and budget-cutter days sit at identical points (the rings mark the budget cutter). Essentials-only days have no comfort by design.

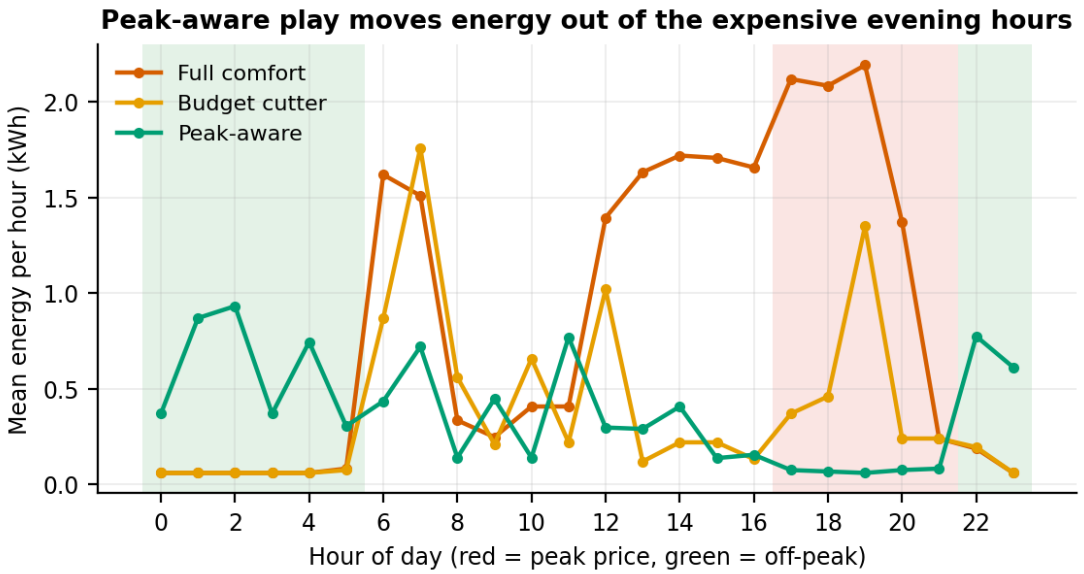


Figure 6. Mean energy per hour on a Medium day. The peak-aware strategy leaves the red evening hours almost empty and uses the green night hours.

Timing is a free saving. At the same energy and comfort, the peak-aware day costs 1.10 against 1.75 for the budget cutter on a Medium day, which is 37% less. Across all 144 paired runs (same level, event and seed) the mean saving is 31.5% (95% interval 30.0 to 33.0) and the score is 60 points higher. The share of energy used at peak falls from 29% to 4% and the off-peak share rises from 7% to 54%. The optimizer, which starts from the peak-aware plan, reaches a score of 755 against 753, so simple rules capture almost everything the search finds.

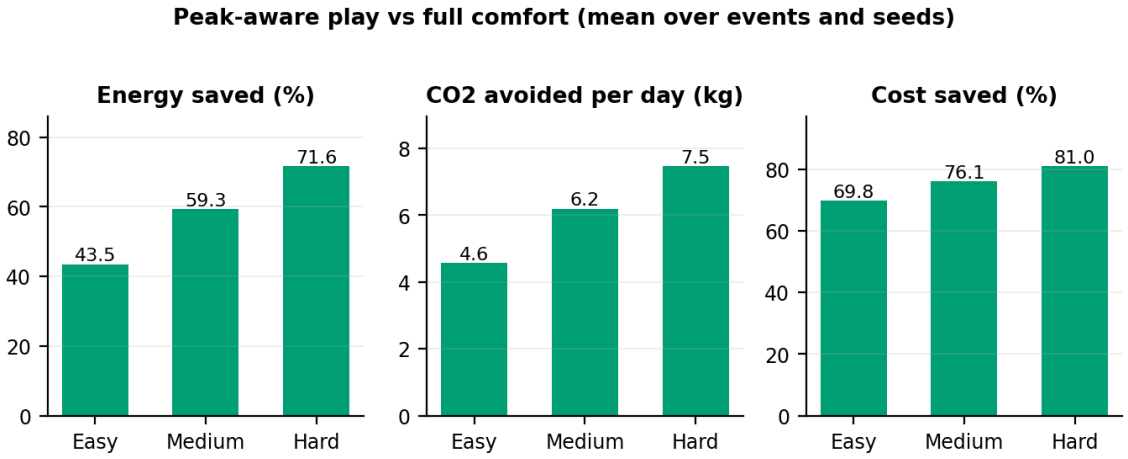


Figure 7. Peak-aware play against a full-comfort day, averaged over the six event settings and eight seeds. Most of the energy saving comes from the budget forcing cuts; the cost saving also includes timing.

6. The sustainable solution

The proposal is the game itself, used as a teaching and awareness tool, together with the habits the data points to. The habits, in order of effect:

- Decide the air conditioner first. It is 56% of the day's energy, and one hour less saves 1.5 kWh.
- Treat the water heater and iron as scheduled jobs. Together they are 14% of the day and can move to the night at no loss of comfort. The cooktop (11%) is tied to meal times, so plan around it.
- Run flexible loads in the off-peak hours. On normal days, across the three levels, this cut cost by 34% on average (pairs with the same level and seed) with energy and comfort unchanged.
- Leave the small loads alone until the large ones are planned.

For a sense of scale: moving from our full-comfort pattern to the budget-meeting pattern avoids about 6.2 kg of CO_2 per Medium day, or roughly 2.3 tonnes over a year if every day looked like the model day. Real days do not, and few households run everything at full comfort, so read this as an upper bound for the illustration and not a forecast. The cost saving from timing carries no CO_2 benefit in this model,

because it uses a single emission factor for every hour. On a real grid the cleanest hours may differ from the cheapest ones.

How it would be used. In a classroom, each student plays a four-minute round, compares scores, and discusses why a heatwave day is harder than a normal one. A teacher can collect the exported CSV files and run the cleaning and analysis scripts to see what a class actually does. The game runs from one command and needs no internet connection after installation. Putting it online (for example on Streamlit Community Cloud) is possible but not done or tested.

7. Testing and robustness

The project has 58 automated tests covering the energy, CO_2, cost and score calculations, the events, the data pipeline (including broken rows and a browser export), the tips, a smoke test of the app, and a check that the browser's scoring matches Python's on random days for every level and event. A scripted browser run played 90 full days through the real interface, each confirmed by the Python engine, with no mismatches or script errors. We also replayed the Python peak-aware schedule through the interface: it scored 753 (Gold) with the same energy and cost as Python.

Sensitivity. Three game values sit below their published ranges. We set the electric cooktop, water heater and iron to the lowest published values (2000 W, 4000 W and 1100 W) and re-ran the scripted strategies on a normal day. A full-comfort day then uses 25.0 kWh (up 17%), so absolute kWh figures are approximate. The ranking of the strategies by score is unchanged (peak-aware, budget cutter, essentials only, casual, full comfort, from best to worst) and peak-aware still costs 28.5% less than the budget cutter. The relative findings hold; the absolute numbers should be read loosely.

8. Limitations

- There are no human players in the data. The scripted strategies are ours, and a real person might behave quite differently.
- Three appliance values are below published ranges, as above. The tariff (a 1:2:4 pattern) is invented, not a real utility's, and one emission factor is used for every hour and region.
- A day is one isolated 24 hours. There is no weather forecast, solar panel or battery, and no carry-over between days.
- Par scores come from a random search and are approximate. The optimizer starts from our own scripted plan, so a cleverer human plan could beat par.
- The emission factor of 0.445 kg/kWh comes from search results quoting the IEA report [1]. The page itself blocks automated access, so we have not read it. It should be confirmed in a browser before submission. The other facts shown in the game were read on their pages (EPA [2], RunWatts [4]).
- Layout was checked in a desktop browser pane and at phone width, not on real phones or a slow college computer. The game has not been deployed online.

9. Next steps

Collect at least twenty real runs, ideally from students who have not seen the game, and re-run the analysis. The play-test kit in the repository has the observation sheet. After that, add region presets for tariff and emission factor, and a solar and battery option with a cloudy-day event.

References

[1] IEA, Electricity 2025, Emissions (2024 global average carbon intensity; not read directly, see section 8). https://www.iea.org/reports/electricity-2025/emissions
[2] U.S. EPA, Greenhouse Gas Equivalencies Calculator, calculations and references (3.93e-4 metric tons CO_2e per mile; read 10 Nov 2026). https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator-calculations-and-references
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[5] Enphase, How many watts do common home appliances use? (updated 18 Nov 2025). https://enphase.com/blog/homeowners/how-many-watts-do-common-home-appliances-use
[6] Source code, data and documents: https://github.com/Pranav0774/energy-conservation-game